

The d- and f-Block Elements

Chapter 4 · All questions with full solutions



Part 1 · Examples

Q

4.1

Transition
element vs
non-
transition
element



On what ground can you say that scandium ($Z = 21$) is a transition element but zinc ($Z = 30$) is not?

Conclusion

Scandium has incompletely filled 3 d orbitals in its ground state ($3d^1$). So it is regarded as a transition element. Zinc, on the other hand, has completely filled d orbitals ($3d^{10}$) in its ground state as well as in its oxidised state. So it is not regarded as a transition element.

Q

4.2

High
enthalpy of
atomisation



Why do the transition elements exhibit higher enthalpies of atomisation?

Conclusion

Transition elements have a large number of unpaired electrons in their atoms. This gives stronger interatomic interaction and hence stronger bonding between atoms.
As a result, they have higher enthalpies of atomisation.

Q

4.3

Element
without
variable
oxidation
states

(())

Name a transition element which does not exhibit variable oxidation states.

Conclusion

Scandium ($Z = 21$) does not exhibit variable oxidation states.

Q

4.4

Cr^{2+}
reducing vs
 Mn^{3+}
oxidising



Why is Cr^{2+} reducing and Mn^{3+} oxidising when both have d^4 configuration?

Conclusion

Cr^{2+} is reducing because its configuration changes from d^4 to d^3 . The latter has a half-filled t_{2g} level (see Unit 5). On the other hand, the change from Mn^{3+} to Mn^{2+} gives the half-filled (d^5) configuration, which has extra stability.

Q

4.5

Oxidising
power trend
of oxoanions

How would you account for the increasing oxidising power in the series $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$?

Conclusion

This is due to the increasing stability of the lower species to which they are reduced.

Q

4.6

Irregular
 M^{2+} / M
electrode
potentials

For the first row transition metals the E° values are:

E°	V	Cr	Mn	Fe	Co	Ni	Cu
(M^{2+} / M)	-1.18	-0.91	-1.18	-0.44	-0.28	-0.25	+0.34

Explain the irregularity in the above values.

Conclusion

The $E^\circ(\text{M}^{2+} / \text{M})$ values are not regular. This can be explained by the irregular variation of ionisation enthalpies ($\Delta_i H_1 + \Delta_i H_2$). It can also be explained by the sublimation enthalpies, which are relatively much less for manganese and vanadium.

Q

4.7

 $\text{Mn}^{3+} / \text{Mn}^{2+}$
electrode
potential
anomaly

()

Why is the E° value for the $\text{Mn}^{3+} / \text{Mn}^{2+}$ couple much more positive than that for $\text{Cr}^{3+} / \text{Cr}^{2+}$ or $\text{Fe}^{3+} / \text{Fe}^{2+}$? Explain.

Conclusion

This is mainly because Mn has a much larger third ionisation energy (the required change here is d^5 to d^4). This also explains why the +3 state of Mn is of little importance.

Q

4.8

Magnetic
moment
calculation

Calculate the magnetic moment of a divalent ion in aqueous solution if its atomic number is 25.

Given

With atomic number 25, the divalent ion in aqueous solution will have d^5 configuration (five unpaired electrons).

Key formula

The magnetic moment, μ is

$$\mu = \sqrt{5(5+2)} = 5.92 \text{ BM}$$

$$\mu = 5.92 \text{ BM}$$

///

Q

4.9

Disproportionation of oxidation state



What is meant by 'disproportionation' of an oxidation state? Give an example.

Given

Disproportionation happens when a particular oxidation state becomes less stable than other oxidation states, one lower and one higher.

For example, manganese (VI) becomes unstable relative to manganese(VII) and manganese (IV) in acidic solution.

Key formula



Q

4.10

Lanthanoid with +4 oxidation state



Name a member of the lanthanoid series which is well known to exhibit +4 oxidation state.

Conclusion

Cerium (Z = 58)

Q

1

Transition
element
definition

(())

Silver atom has completely filled d orbitals ($4d^{10}$) in its ground state. How can you say that it is a transition element?

Conclusion

Ag has a completely filled 4 d orbital ($4d^{10} 5s^1$) in its ground state. Silver shows two oxidation states, +1 and +2. In the +1 state, one electron is removed from the s-orbital. In the +2 state, an electron is removed from the d-orbital. This makes the d-orbital incomplete ($4d^9$). So, silver is a transition element.

Q

2

Enthalpy of
atomisation
of zinc

()

In the series Sc(Z = 21) to Zn(Z = 30), the enthalpy of atomisation of zinc is the lowest, i.e., 126 kJ mol^{-1} . Why?

Conclusion

The enthalpy of atomization depends on how strong the metallic bonding is. The stronger the metallic bonding, the higher the enthalpy of atomization. All transition metals, except Zn (electronic configuration: $3d^{10} 4s^2$), have some unpaired electrons. These unpaired electrons give stronger metallic bonding. Zn has no such unpaired electrons, so its inter-atomic bonding is the weakest. This is why Zn has the lowest enthalpy of atomization.

Q

3

Number of
oxidation
states in 3d
series

()

Which of the 3 d series of the transition metals exhibits the largest number of oxidation states and why?

Key formula

$\text{Mn}(Z = 25) = 3d^5 4s^2$

()

Conclusion

Mn has the maximum number of unpaired electrons present in the d-subshell (5 electrons). Hence, Mn exhibits the largest number of oxidation states, ranging from +2 to +7.

Q

4

Positive
electrode
potential of
copper

()

The $E^\ominus(\text{M}^{2+} / \text{M})$ value for copper is positive (+0.34V). What is possible reason for this? (Hint: consider its high $\Delta_a H^\ominus$ and low $\Delta_{\text{hyd}} H^\ominus$)

Given

The $E^\ominus(\text{M}^{2+} / \text{M})$ value of a metal depends on the energy changes involved in the following:

1. Sublimation: The energy required for converting one mole of an atom from the solid state to the gaseous state.

Key formula

$\text{M}_{(\text{s})} \rightarrow \text{M}_{(\text{g})} \Delta_s H$ (Sublimation energy)

()

Substitute

2. Ionization: The energy required to take out electrons from one mole of atoms in the gaseous state.

Key formula

$\text{M}_{(\text{g})} \rightarrow \text{M}_{(\text{g})}^{2+} \Delta_i H$ (Ionization energy)

**Substitute**

3. Hydration: The energy released when one mole of ions are hydrated.

Key formula**Conclusion**

Copper has a high energy of atomization and low hydration energy. So, the $E^\ominus(M^{2+} / M)$ value for copper is positive.

Q**5**

**Irregular
ionisation
enthalpies**

(())

How would you account for the irregular variation of ionisation enthalpies (first and second) in the first series of the transition elements?

Conclusion

Ionization enthalpies increase across the series because the inner d-orbitals keep filling up. Some configurations, such as d^0 , d^5 , d^{10} , are extra stable. Elements with these configurations have very high ionization enthalpies.

For the first ionization energy, Cr has a low value. This is because after losing one electron, Cr reaches the stable configuration ($3d^5$). Zn, on the other hand, has an exceptionally high first ionization energy, because an electron has to be removed from its stable, fully-filled orbitals ($3d^{10} 4s^2$).

Second ionization energies are higher than first ionization energies, since it is harder to remove an electron once one electron is already gone. Cr and Cu also have exceptionally high second ionization energies, because after losing the first electron, they reach the stable configurations ($Cr^+ : 3d^5$, and $Cu^+ : 3d^{10}$). Removing one more electron from this stable configuration needs a lot of energy.

Q**6**

**Highest
oxidation
state in
oxides / fluo
rides**



Why is the highest oxidation state of a metal exhibited in its oxide or fluoride only?

Conclusion

Both oxide and fluoride ions are highly electronegative and have a very small size. Due to these properties, they are able to oxidize the metal to its highest oxidation state.

Q**7**

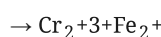
**Comparing
reducing
agents Cr^{2+}
and Fe^{2+}**



Which is a stronger reducing agent Cr^{2+} or Fe^{2+} and why ?

Conclusion

The following reactions are involved when Cr^{2+} and Fe^{2+} act as reducing agents.



$E^\ominus Cr^{3+} / Cr^{2+}$ The value is $-0.41 V$ and $E^\ominus Fe^{3+} / Fe^{2+}$ is $+0.77 V$. This means that Cr^{2+} can

be easily oxidized to Cr^{3+} , but Fe^{2+} does not get oxidized to Fe^{3+} easily. Therefore, Cr^{2+} is a better reducing agent than Fe^{3+} .

Q

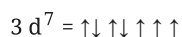
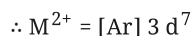
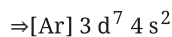
8

Spin-only
magnetic
moment
calculation

Calculate the 'spin only' magnetic moment of $M^{2+}_{(aq)}$ ion ($Z = 27$).

Given

$$Z = 27$$



i.e., 3 unpaired electrons

Substitute

$$\therefore n = 3$$

$$\Rightarrow \sqrt{n(n+2)} = \mu$$

$$\Rightarrow \sqrt{3(3+2)} = \mu$$

$$\Rightarrow \sqrt{15} = \mu$$

$$\mu \approx 4 \text{ BM}$$

Q

9

Stability of
 Cu^+ in
aqueous
solution

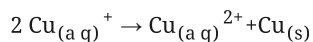
()

Explain why Cu^+ ion is not stable in aqueous solutions?

Given

In an aqueous medium, Cu^{2+} is more stable than Cu^+ . Energy is needed to remove one electron from Cu^+ to Cu^{2+} , but the high hydration energy of Cu^{2+} makes up for it. So, Cu^+ ion is unstable in aqueous solution. It disproportionates to give Cu^{2+} and Cu .

Key formula



Q

10

Actinoid vs
lanthanoid
contraction

Actinoid contraction is greater from element to element than lanthanoid contraction.

Why?

Conclusion

In actinoids, 5 f orbitals are filled. These 5 f orbitals have a poorer shielding effect than 4 f orbitals (in lanthanoids). So, the effective nuclear charge on valence-shell electrons is much higher in actinoids than in lanthanoids. This is why the size contraction is greater in actinoids than in lanthanoids.

Q

1

Electronic
configuration
s of
transition
ions



Write down the electronic configuration of:

Part (i)

Cr^{3+}

Conclusion

$\text{Cr}^{3+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$

Or, $[\text{Ar}]^{18} 3d^3$

Part (ii)

Pm^{3+}

Conclusion

$\text{Pm}^{3+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^4$

Or, $[\text{Xe}]^{54} 3d^3$

Part (iii)

Cu^+

Conclusion

$\text{Cu}^+: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

Or, $[\text{Ar}]^{18} 3d^{10}$

Part (iv)

Ce^{4+}

Conclusion

$\text{Ce}^{4+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6$

Or, $[\text{Xe}]^{54}$

Part (v)

Co^{2+}

Conclusion

$\text{Co}^{2+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$

Or, $[\text{Ar}]^{18} 3d^7$

Part (vi)

Lu^{2+}

Conclusion

$\text{Lu}^{2+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^{14} 5d^1$

Or, $[\text{Xe}]^{54} 2f^{14} 3d^3$

Part (vii)

Mn^{2+}

Conclusion

$\text{Mn}^{2+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Or, $[\text{Ar}]^{18} 3d^5$

Part (viii)

Th^{4+}

Conclusion

$\text{Th}^{4+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 6s^2 6p^6$

Or, $[\text{Rn}]^{86}$

Q

2

Mn²⁺ vs Fe²⁺
stability to
oxidation

Why are Mn²⁺ compounds more stable than Fe²⁺ towards oxidation to their +3 state?

Conclusion

Electronic configuration of Mn²⁺ is [Ar]¹⁸ 3 d⁵. Electronic configuration of Fe²⁺ is [Ar]¹⁸ 3 d⁶.

Half-filled and fully-filled orbitals are more stable. So Mn in the (+2) state has a stable d⁵ configuration. This is why Mn²⁺ resists oxidation to Mn³⁺.

Fe²⁺ has 3 d⁶ configuration. By losing one electron, it changes to a more stable 3 d⁵ configuration. So Fe²⁺ easily gets oxidized to the Fe⁺³ oxidation state.

Q

3

Stability
trend of +2
oxidation
state

()

Explain briefly how +2 state becomes more and more stable in the first half of the first row transition elements with increasing atomic number?

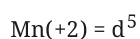
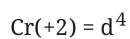
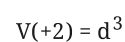
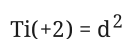
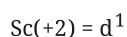
Given

The oxidation states displayed by the first half of the first row of transition metals are given in the table below.

Oxidation state	Sc	Ti	V	Cr	Mn
		+2	+2	+2	+2
	+3	+3	+3	+3	+3
		+4	+4	+4	+4
			+5	+5	+6
				+6	+7

Conclusion

We can see that all the metals except Sc show the +2 oxidation state. Also, moving from Sc to Mn, the atomic number increases from 21 to 25. So the number of electrons in the 3 d-orbital also increases from 1 to 5.



These metals reach the +2 oxidation state by losing two 4 s electrons. The number of d electrons in the (+2) state also increases from Ti(+2) to Mn(+2), so the stability of the +2 state increases, as the d-orbital becomes more and more half-filled. Mn (+2) has d⁵ electrons — a half-filled d shell, which is highly stable.

Q

4

Electronic configuration and oxidation state stability



To what extent do the electronic configurations decide the stability of oxidation states in the first series of the transition elements? Illustrate your answer with examples.

Conclusion

The elements in the first half of the transition series show many oxidation states. Mn shows the maximum number, from +2 to +7. The stability of the +2 oxidation state increases as the atomic number increases, because more electrons are filling the d-orbital.

Sc, however, does not show a +2 oxidation state. Its electronic configuration is $4s^2 3d^1$. It loses all three electrons to form Sc^{3+} . The +3 state of Sc is very stable because, by losing all three electrons, it gets the stable noble gas configuration [Ar]. Ti (+4) and V(+5) are very stable for the same reason.

For Mn, the +2 state is very stable because after losing two electrons, its d-orbital is exactly half-filled, $[Ar] 3d^5$.

Q

5

Stable oxidation states from d-configuration



What may be the stable oxidation state of the transition element with the following d electron configurations in the ground state of their atoms : $3d^3$, $3d^5$, $3d^8$ and $3d^4$?

	Electronic configuration in ground state	Stable oxidation states
(i)	$3d^3$ (Vanadium)	+2, +3, +4 and +5
(ii)	$3d^5$ (Chromium)	+3, +4, +6
	:-:- :-:-	:-:-
(iii)	$3d^5$ (Manganese)	+2, +4, +6, +7
(iv)	$3d^8$ (Cobalt)	+2, +3
(v)	$3d^4$	There is no $3d^4$ configuration in ground state.

Q

6

Oxometal anions in highest group oxidation state



Name the oxometal anions of the first series of the transition metals in which the metal exhibits the oxidation state equal to its group number.

Conclusion

(i) Vanadate, VO_3^-

Oxidation state of V is +5.

(ii) Chromate, CrO_4^{2-}

Oxidation state of Cr is +6.

(iii) Permanganate, MnO_4^-

Oxidation state of Mn is +7.

Q

7

Lanthanoid contraction and its consequences

((

What is lanthanoid contraction? What are the consequences of lanthanoid contraction?

Conclusion

As we move along the lanthanoid series, the atomic number increases by one each time.

So the number of electrons and protons in the atom also increases by one.

Since the new electrons are added to the same shell, the effective nuclear charge increases.

This is because the increase in nuclear attraction from the added proton is greater than the increase in repulsion from the added electron.

Also, as the atomic number increases, the number of electrons in the 4 f orbital increases too. These 4 f electrons shield poorly. So the effective nuclear charge felt by the outer electrons increases, and the nucleus pulls the outermost electrons in more strongly.

This causes a steady decrease in the size of the lanthanoids as the atomic number increases.

This is called lanthanoid contraction.

Consequences of lanthanoid contraction

- (i) It explains the similarity in the properties of the second and third transition series.
- (ii) It makes it possible to separate the lanthanoids from each other.
- (iii) It causes the basic strength of the lanthanide hydroxides to vary. (Basic strength decreases from $\text{La}(\text{OH})_3$ to $\text{Lu}(\text{OH})_3$.)

Q

8

Characteristics of transition elements



What are the characteristics of the transition elements and why are they called transition elements? Which of the d-block elements may not be regarded as the transition elements?

Conclusion

Transition elements are elements whose atoms or ions (in a stable oxidation state) have a partially filled d-orbital. These elements lie in the d-block. Their properties are a transition between the s-block and the p-block. That is why they are called transition elements.

Zn, Cd, and Hg are not classified as transition elements because they have a completely filled d-subshell.

Q

9

Electronic configuration : transition vs non-transition



In what way is the electronic configuration of the transition elements different from that of the non transition elements?

Conclusion

Transition metals have a partially filled d-orbital. So the electronic configuration of transition elements is $(n-1) d^{1-10} n s^{0-2}$.

Non-transition elements either have no d-orbital or have a fully filled d-orbital. So the electronic configuration of non-transition elements is $n s^{1-2}$ or $n s^2 n p_{1-6}$.

Q

10

Oxidation
states
exhibited by
lanthanoids



What are the different oxidation states exhibited by the lanthanoids?

Conclusion

In the lanthanide series, +3 oxidation state is most common i.e., Ln(III) compounds are predominant. However, +2 and +4 oxidation states can also be found in the solution or in solid compounds.

Q

11

Reasons for
transition
metal
properties

()

Explain giving reasons:

Part (i)

Transition metals and many of their compounds show paramagnetic behaviour.

Conclusion

Transition metals show paramagnetic behaviour.

Paramagnetism happens because of unpaired electrons — each unpaired electron has a magnetic moment from its spin and orbital angular momentum.

But in the first transition series, the orbital angular momentum is quenched.

So the paramagnetism we see comes only from the unpaired electrons.

Part (ii)

The enthalpies of atomisation of the transition metals are high.

Conclusion

Transition elements have a high effective nuclear charge and many valence electrons.

So they form very strong metallic bonds.

This is why the enthalpy of atomisation of transition metals is high.

Part (iii)

The transition metals generally form coloured compounds.

Conclusion

Most complexes of transition metals are coloured. This happens because an electron absorbs radiation from the visible region and jumps from one d-orbital to another.

In the presence of ligands, the d-orbitals split into two sets with different energies. So electrons can move from one set to another. This transition needs only a small amount of energy, which falls in the visible region.

The metal ion absorbs radiation of one particular wavelength and reflects the rest.

This reflected light gives the solution its colour.

Part (iv)

Transition metals and their many compounds act as good catalyst.

Conclusion

The catalytic activity of transition elements can be explained by two facts.

(a) Because they can show variable oxidation states and form complexes, transition metals form unstable intermediate compounds. This gives the reaction a new path with lower activation energy, E_a .

(b) Transition metals also give the reaction a suitable surface to occur on.

Q

12

Interstitial compounds in transition metals



What are interstitial compounds? Why are such compounds well known for transition metals?

Conclusion

Transition metals are large and have many interstitial sites.

They can trap small atoms of other elements — such as H, C, and N — in these interstitial sites of their crystal lattice.

The compounds formed this way are called interstitial compounds.

Q

13

Variable oxidation states comparison



How is the variability in oxidation states of transition metals different from that of the non transition metals? Illustrate with examples.

Conclusion

In transition elements, the oxidation state can vary from +1 up to the highest state, formed by removing all the valence electrons. These oxidation states differ from each other by 1 (Fe^{2+} and Fe^{3+} ; Cu^+ and Cu^{2+}).

In non-transition elements, the oxidation states differ by 2 — for example, +2 and +4 or +3 and +5, etc.

Q

14

Potassium dichromate preparation and pH effect

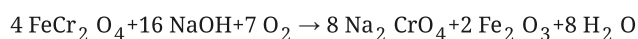


Describe the preparation of potassium dichromate from iron chromite ore. What is the effect of increasing pH on a solution of potassium dichromate?

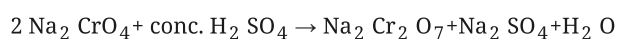
Conclusion

Potassium dichromate is prepared from chromite ore (FeCr_2O_4) in the following steps.

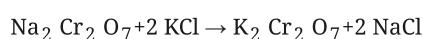
Step (1): Preparation of sodium chromate



Step (2): Conversion of sodium chromate into sodium dichromate



Step(3): Conversion of sodium dichromate to potassium dichromate



Potassium chloride is less soluble than sodium chloride.

So it is obtained as orange coloured crystals, which can be removed by filtration.

The dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$) exists in equilibrium with the chromate ion (CrO_4^{2-})_{ion at pH 4}. Changing the pH can convert one into the other.



Chromate

Hydrogen

Dichromate

(Yellow)

chromate

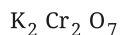
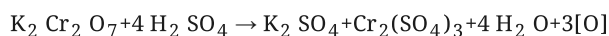
(Orange)

Q**15****Oxidising
action of
potassium
dichromate****(())**

Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with:

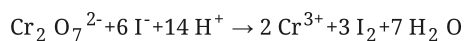
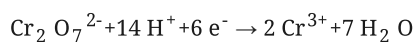
Part (i)**iodide****Given**

$\text{K}_2 \text{Cr}_2 \text{O}_7$ acts as a very strong oxidising agent in the acidic medium.

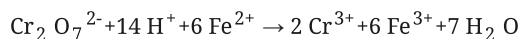
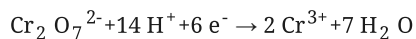


It takes up electrons, gets reduced, and acts as an oxidising agent. Reactions of $\text{K}_2 \text{Cr}_2 \text{O}_7$ with iodide, iron (II) solution, and $\text{H}_2 \text{S}$ are given below.

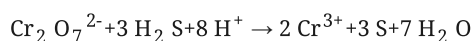
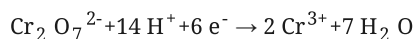
$\text{K}_2 \text{Cr}_2 \text{O}_7$ oxidizes iodide to iodine.

Key formula**Part (ii)****iron(II) solution and****Given**

$\text{K}_2 \text{Cr}_2 \text{O}_7$ oxidizes iron (II) solution to iron (III) solution i.e., ferrous ions to ferric ions.

Key formula**Part (iii)** **$\text{H}_2 \text{S}$** **Given**

$\text{K}_2 \text{Cr}_2 \text{O}_7$ oxidizes $\text{H}_2 \text{S}$ to sulphur.

Key formula

Q

16

Potassium
permanganate
preparation
and
reactions

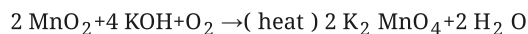


Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with (i) iron(II) ions (ii) SO_2 and (iii) oxalic acid?

Write the ionic equations for the reactions.

Conclusion

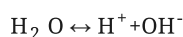
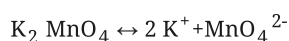
Potassium permanganate is prepared from pyrolusite (MnO_2). The ore is fused with KOH in the presence of atmospheric oxygen or an oxidising agent, such as KNO_3 or KClO_4 , to get K_2MnO_4 .



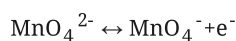
(Green)

The green mass is extracted with water and then oxidised, either electrolytically or by passing chlorine / ozone into the solution.

Electrolytic oxidation

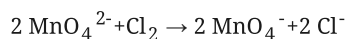
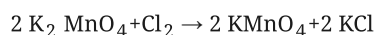


At the anode, manganate ions are oxidised to permanganate ions.

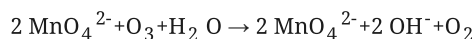
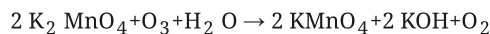


Green Purple

Oxidation by chlorine



Oxidation by ozone



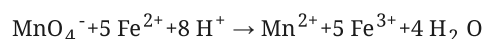
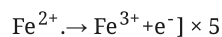
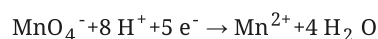
Part (i)

Given

Acidified KMnO_4 solution oxidizes Fe (II) ions to Fe (III) ions i.e., ferrous ions to ferric ions.



Key formula



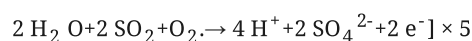
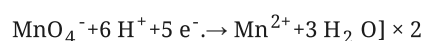
Part (ii)

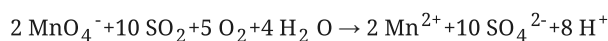
Given

Acidified potassium permanganate oxidizes SO_2 to sulphuric acid.



Key formula





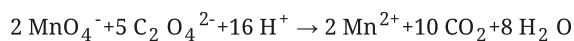
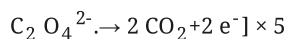
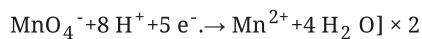
Part (iii)

Given

Acidified potassium permanganate oxidizes oxalic acid to carbon dioxide.



Key formula



Q

17

E-degree values and metal ion stability



For M^{2+} / M and $\text{M}^{3+} / \text{M}^{2+}$ systems the E° values for some metals are as follows:



Use this data to comment upon:

Part (i)

the stability of Fe^{3+} in acid solution as compared to that of Cr^{3+} or Mn^{3+} and

Conclusion

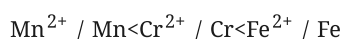
The E° value for $\text{Fe}^{3+} / \text{Fe}^{2+}$ is higher than that for $\text{Cr}^{3+} / \text{Cr}^{2+}$ and lower than that for $\text{Mn}^{3+} / \text{Mn}^{2+}$. So, the reduction of Fe^{3+} to Fe^{2+} is easier than the reduction of Mn^{3+} to Mn^{2+} , but not as easy as the reduction of Cr^{3+} to Cr^{2+} . Hence, Fe^{3+} is more stable than Mn^{3+} , but less stable than Cr^{3+} . These metal ions can be arranged in the increasing order of their stability as: $\text{Mn}^{3+} < \text{Fe}^{3+} < \text{Cr}^{3+}$

Part (ii)

the ease with which iron can be oxidised as compared to a similar process for either chromium or manganese metal.

Conclusion

The reduction potentials for the given pairs increase in the following order.



So, the oxidation of Fe to Fe^{2+} is not as easy as the oxidation of Cr to Cr^{2+} and the oxidation of Mn to Mn^{2+} . Thus, these metals can be arranged in the increasing order of their ability to get oxidised as: $\text{Fe} < \text{Cr} < \text{Mn}$

Q

18

Colour
prediction of
aqueous ions



Predict which of the following will be coloured in aqueous solution? Ti^{3+} , V^{3+} , Cu^+ , Sc^{3+} , Mn^{2+} , Fe^{3+} and Co^{2+} . Give reasons for each.

Given

Only ions with electrons in the d-orbital are coloured. Ions with an empty d-orbital are colourless.

Element	Atomic Number	Ionic State	Electronic configuration in ionic state
Ti	22	Ti^{3+}	$[\text{Ar}] 3d^1$
---	---	---	---
V	23	V^{3+}	$[\text{Ar}] 3d^2$
Cu	29	Cu^+	$[\text{Ar}] 3d^{10}$
Sc	21	Sc^{3+}	$[\text{Ar}]$
Mn	25	Mn^{2+}	$[\text{Ar}] 3d^5$
Fe	26	Fe^{3+}	$[\text{Ar}] 3d^5$
Co	27	Co^{2+}	$[\text{Ar}] 3d^7$

Conclusion

From the table, we can see that only Sc^{3+} has an empty d-orbital. All the other ions, except Sc^{3+} , are coloured in aqueous solution because of d-d transitions.

Q

19

Stability of
+2 oxidation
state
comparison

(())

Compare the stability of +2 oxidation state for the elements of the first transition series.

Sc	+3						
Ti	+1	+2	+3	+4			
V	+1	+2	+3	+4	+5		
Cr	+1	+2	+3	+4	+5	+6	
Mn	+1	+2	+3	+4	+5	+6	+7
Fe	+1	+2	+3	+4	+5	+6	
Co	+1	+2	+3	+4	+5		
---	---	---	---	---	---	---	---
Ni	+1	+2	+3	+4			
Cu	+1	+2	+3				
Zn		+2					

Conclusion

From the table, we can see that Mn shows the maximum number of oxidation states, varying from +2 to +7. The number of oxidation states increases from Sc to Mn, then decreases from Mn to Zn, because the number of available unpaired electrons decreases. The +2 oxidation state becomes more stable moving from top to bottom. This is because it becomes harder and harder to remove the third electron from the d-orbital as we move from top to bottom.

Q

20

Actinoids vs
lanthanoids
comparison

Compare the chemistry of actinoids with that of the lanthanoids with special reference to:

Part (i)

electronic configuration

Conclusion

The general electronic configuration is $[\text{Xe}]^{54} 4f^{0-14} 5d^{0-1} 6s^2$ for lanthanoids and $[\text{Rn}]^{86} 5f^{1-14} 6d^{0-1} 7s^2$ for actinoids. Unlike 4 f orbitals, the 5 f orbitals are not buried as deeply, so they take part in bonding more.

Part (ii)

atomic and ionic sizes and

Conclusion

Like lanthanoids, actinoids also show actinoid contraction — an overall decrease in atomic and ionic radii. This contraction is greater, because of the poor shielding effect of the 5 f orbitals.

Part (iii)

oxidation state

Conclusion

The main oxidation state of lanthanoids is +3. We also sometimes see +2 and +4, because of the extra stability of fully-filled and half-filled orbitals.

Actinoids show a wider range of oxidation states, because the 5 f, 6 d and 7 s levels have comparable energies. But +3 is still the main oxidation state for actinoids too. Like lanthanoids, actinoids form more compounds in the +3 state than in the +4 state.

Part (iv)

chemical reactivity.

Conclusion

In the lanthanide series, the earlier members are more reactive — their reactivity is comparable to Ca.

As the atomic number increases, the lanthanides start behaving more like Al.

Actinoids, on the other hand, are highly reactive metals, especially when finely divided.

Added to boiling water, they give a mixture of oxide and hydride.

They combine with most non-metals at moderate temperatures.

Alkalies have no effect on them.

Among acids, only nitric acid affects them slightly, because it forms a protective oxide layer.

Q

21

Reasons for
transition
metal ion
behaviour

How would you account for the following:

Part (i)

Of the d^4 species, Cr^{2+} is strongly reducing while manganese(III) is strongly oxidising.

Conclusion

Cr^{2+} is strongly reducing. It has a d^4 configuration. As a reducing agent, it gets oxidised to Cr^{3+} (electronic configuration d^3). This d^3 configuration can also be written as t_{2g}^3 configuration, which is more stable.

Mn^{3+} (d^4), on the other hand, acts as an oxidizing agent and gets reduced to Mn^{2+} (d^5). This has an exactly half-filled d-orbital, so it is highly stable.

Part (ii)

Cobalt(II) is stable in aqueous solution but in the presence of complexing reagents it is easily oxidised.

Conclusion

Co(II) is stable in aqueous solution. But in the presence of strong field complexing reagents, it is oxidised to Co (III). The 3rd ionization energy for Co is high. However, the larger crystal field stabilization energy (CFSE) released by strong field ligands is enough to overcome this ionization energy.

Part (iii)

The d¹ configuration is very unstable in ions.

Conclusion

Ions with a d¹ configuration tend to lose one more electron to reach the stable d⁰ configuration. Also, the hydration or lattice energy is more than enough to remove the one electron present in the d-orbital of these ions. So they act as reducing agents.

Q

22

Disproportionation
reaction
examples



What is meant by 'disproportionation'? Give two examples of disproportionation reaction in aqueous solution.

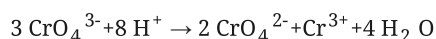
Given

Sometimes a relatively less stable oxidation state undergoes a reaction in which it is oxidised and reduced at the same time.

This is called disproportionation.

For example,

Key formula



Cr(V) Cr(VI) Cr(III)

Substitute

Here, Cr(V) is oxidized to Cr(VI) and reduced to Cr(III).

Key formula



Mn(VI) Mn(VII) Mn(IV)

Conclusion

Here, Mn(VI) is oxidized to Mn(VII) and reduced to Mn(IV).

Q

23

Metal
favouring +1
oxidation
state

(())

Which metal in the first series of transition metals exhibits +1 oxidation state most frequently and why?

Conclusion

In the first transition series, Cu very often shows the +1 oxidation state. This is because Cu(+1) has an electronic configuration of [Ar] 3d¹⁰. Its completely filled d-orbital makes it highly stable.

Q

24

Unpaired
electrons in
gaseous ions

Calculate the number of unpaired electrons in the following gaseous ions: Mn^{3+} , Cr^{3+} , V^{3+} and Ti^{3+} . Which one of these is the most stable in aqueous solution?

	Gaseous ions	Number of unpaired electrons
(i)	$\text{Mn}^{3+}, [\text{Ar}] 3d^4$	4
(ii)	$\text{Cr}^{3+}, [\text{Ar}] 3d^3$	3
(iii)	$\text{V}^{3+}, [\text{Ar}] 3d^2$	2
(vi)	$\text{Ti}^{3+}, [\text{Ar}] 3d^1$	1

Conclusion

Cr^{3+} is the most stable in aqueous solutions owing to a t_{2g}^3 configuration.

Q

25

Transition
metal oxide
and
oxoanion
trends

Give examples and suggest reasons for the following features of the transition metal chemistry:

Part (i)

The lowest oxide of transition metal is basic, the highest is amphoteric / acidic.

Conclusion

In a lower oxide of a transition metal, the metal atom has a low oxidation state.

So some of its valence electrons are not used in bonding, and it can donate electrons — behaving as a base.

In a higher oxide, the metal atom has a high oxidation state.

Here the valence electrons are used up in bonding, so they are unavailable, and the effective nuclear charge is also high.

So it can accept electrons — behaving as an acid.

For example, $\text{Mn}^{\text{II}}\text{O}$ is basic and $\text{Mn}_2^{\text{VII}}\text{O}_7$ is acidic.

Part (ii)

A transition metal exhibits highest oxidation state in oxides and fluorides.

Conclusion

Oxygen and fluorine are strong oxidising agents, because they are highly electronegative and small in size. So they bring out the highest oxidation states in transition metals — which is why a transition metal shows its highest oxidation states in oxides and fluorides. For example, in OsF_6 and V_2O_5 , the oxidation states of Os and V are +6 and +5 respectively.

Part (iii)

The highest oxidation state is exhibited in oxoanions of a metal.

Conclusion

Oxygen is a strong oxidising agent because of its high electronegativity and small size.

So, oxo-anions of a metal have the highest oxidation state. For example, in MnO_4^- , the oxidation state of Mn is +7.



Indicate the steps in the preparation of:

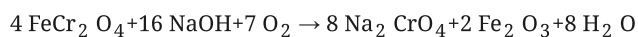
Part (i)

K₂Cr₂O₇ from chromite ore.

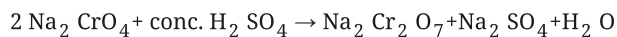
Conclusion

Potassium dichromate (K₂Cr₂O₇) is prepared from chromite ore (FeCr₂O₄) in the following steps.

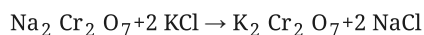
Step (1): Preparation of sodium chromate



Step (2): Conversion of sodium chromate into sodium dichromate



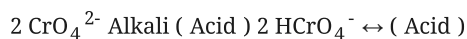
Step(3): Conversion of sodium dichromate to potassium dichromate



Potassium chloride is less soluble than sodium chloride.

So it is obtained as orange coloured crystals, which can be removed by filtration.

The dichromate ion (Cr₂O₇²⁻) exists in equilibrium with the chromate ion (CrO₄²⁻)_{ion} at pH 4. Changing the pH can convert one into the other.



Chromate Hydrogen

(Yellow) chromate

(Yellow Dichromate

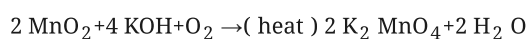
(Orange)

Part (ii)

KMnO₄ from pyrolusite ore.

Conclusion

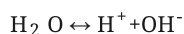
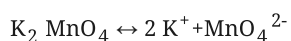
Potassium permanganate (KMnO₄) is prepared from pyrolusite (MnO₂). The ore is fused with KOH in the presence of atmospheric oxygen or an oxidising agent, such as KNO₃ or KClO₄, to get K₂MnO₄.



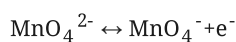
(Green)

The green mass is extracted with water and then oxidised, either electrolytically or by passing chlorine / ozone into the solution.

Electrolytic oxidation

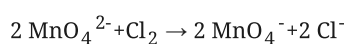
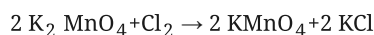


At the anode, manganate ions are oxidised to permanganate ions.

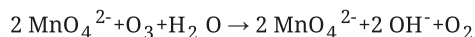
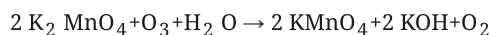


Green Purple

Oxidation by chlorine



Oxidation by ozone



Q

27

Alloys
containing
lanthanoid
metals

((

What are alloys? Name an important alloy which contains some of the lanthanoid metals.

Mention its uses.

Conclusion

An alloy is a solid solution of two or more elements in a metallic matrix.

It can be either a partial solid solution or a complete solid solution.

Alloys usually have different physical properties from those of the component elements.

An important lanthanoid alloy is Mischmetal.

It contains lanthanoids (94-95%), iron (5%), and traces of S, C, Si, Ca, and Al.

Uses

(1) Mischmetal is used in cigarettes and gas lighters.

(2) It is used in flame throwing tanks.

(3) It is used in tracer bullets and shells.

Q

28

Identifying
inner
transition
elements

✧

What are inner transition elements? Decide which of the following atomic numbers are the atomic numbers of the inner transition elements : 29, 59, 74, 95, 102, 104.

Conclusion

Inner transition metals are elements whose last electron enters the f-orbital.

Elements in which the 4 f and the 5 f orbitals are progressively filled are called f-block elements. Among the given atomic numbers, 59, 95, and 102 belong to the inner transition elements.

Q

29

Actinoid
oxidation
state
complexity

✧

The chemistry of the actinoid elements is not so smooth as that of the lanthanoids.

Justify this statement by giving some examples from the oxidation state of these elements.

Conclusion

Lanthanoids mainly show three oxidation states (+2, +3, +4). Of these, +3 is the most common. Lanthanoids show only a limited number of oxidation states because the energy difference between the 4 f, 5 d and 6 s orbitals is quite large.

The energy difference between the 5 f, 6 d and 7 s orbitals, on the other hand, is very small. So actinoids show a large number of oxidation states. For example, uranium and plutonium show +3, +4, +5, and +6 oxidation states, while neptunium shows +3, +4, +5, and +7. The most common oxidation state for actinoids is also +3.

Q

30

Last actinoid
element's
configuration

Which is the last element in the series of the actinoids? Write the electronic configuration of this element.

Comment on the possible oxidation state of this element.

Conclusion

The last element in the actinoid series is lawrencium, Lr. Its atomic number is 103, and its electronic configuration is $[\text{Rn}] 5f^{14} 6d^1 7s^2$. Its most common oxidation state is +3, because losing 3 electrons gives it the stable f^{14} configuration.

Q

31

Ce³⁺
configuration
and magnetic
moment

(())

Use Hund's rule to derive the electronic configuration of Ce³⁺ ion, and calculate its magnetic moment on the basis of 'spin-only' formula.

Given

Ce: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^1 5d^1 6s^2$



Key formula

Magnetic moment can be calculated as:

$$\mu = \sqrt{n(n+2)}$$

Where,

n = number of unpaired electrons



Substitute

In Ce, $n = 2$

Therefore, $\mu = \sqrt{2(2+2)}$

$$= \sqrt{2 \times 4}$$

$$= \sqrt{8}$$

$$= 2\sqrt{2}$$

$$= 2.828 \text{ BM}$$

Q

32

Lanthanoids
with +4 and
+2 oxidation
states

Name the members of the lanthanoid series which exhibit +4 oxidation states and those which exhibit +2 oxidation states.

Try to correlate this type of behaviour with the electronic configurations of these elements.

Given

The lanthanides that show +2 and +4 states are listed in the table below.

The atomic numbers of the elements are given in parentheses.



+2	+4
Nd (60)	Ce (58)
Sm (62)	Pr (59)
Eu (63)	Nd (60)
Tm (69)	Tb (65)
Yb (70)	Dy (66)

Conclusion

Ce forms Ce^{4+} and reaches the stable electronic configuration $[\text{Xe}]$.

Tb forms Tb^{4+} and reaches the stable electronic configuration $[\text{Xe}] 4f^7$.

Eu forms Eu^{2+} and reaches the stable electronic configuration $[\text{Xe}] 4f^7$.

Yb forms Yb^{2+} and reaches the stable electronic configuration $[\text{Xe}] 4f^{14}$.

Q**33**

Actinoid vs
lanthanoid
chemistry
comparison



Compare the chemistry of the actinoids with that of lanthanoids with reference to: (i) electronic configuration (ii) oxidation states and (iii) chemical reactivity.

Conclusion

Electronic configuration

The general electronic configuration is $[\text{Xe}]^{54} 4f^{0-14} 5d^{0-1} 6s^2$ for lanthanoids and $[\text{Rn}]^{86} 5f^{1-14} 6d^{0-1} 7s^2$ for actinoids. Unlike 4f orbitals, the 5f orbitals are not buried as deeply, so they take part in bonding more.

Oxidation states

The main oxidation state of lanthanoids is +3. We also sometimes see +2 and +4 oxidation states, because of the extra stability of fully-filled and half-filled orbitals. Actinoids show a wider range of oxidation states, because the 5f, 6d and 7s levels have comparable energies. But +3 is still the main oxidation state for actinoids too. Like lanthanoids, actinoids form more compounds in the +3 state than in the +4 state.

Chemical reactivity

In the lanthanide series, the earlier members are more reactive — their reactivity is comparable to Ca.

As the atomic number increases, the lanthanides start behaving more like Al.

Actinoids, on the other hand, are highly reactive metals, especially when finely divided.

Added to boiling water, they give a mixture of oxide and hydride.

They combine with most non-metals at moderate temperatures.

Alkalies have no effect on them.

Among acids, only nitric acid affects them slightly, because it forms a protective oxide layer.

Q**34**

Electronic
configuration
s of given
atomic
numbers



Write the electronic configurations of the elements with the atomic numbers 61, 91, 101, and 109.

Atomic number	Electronic configuration
61	$[\text{Xe}]^{54} 4f^5 5d^0 6s^2$
91	$[\text{Rn}]^{86} 5f^2 6d^1 7s^2$
---	---
101	$[\text{Rn}]^{86} 5f^{13} 5d^0 7s^2$
109	$[\text{Rn}]^{86} 5f^{14} 6d^7 7s^2$

Compare the general characteristics of the first series of the transition metals with those of the second and third series metals in the respective vertical columns.

Give special emphasis on the following points:

(i) electronic configurations (ii) oxidation states (iii) ionisation enthalpies and (iv) atomic sizes.

Part (i)

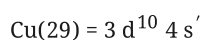
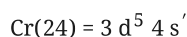
Conclusion

In the 1st, 2nd and 3rd transition series, the 3 d, 4 d and 5 d orbitals are filled, respectively.

Elements in the same vertical column generally have similar electronic configurations.

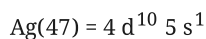
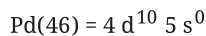
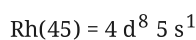
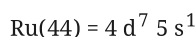
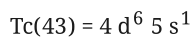
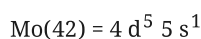
But there are exceptions.

In the first transition series, two elements show unusual electronic configurations:



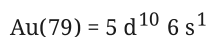
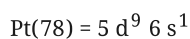
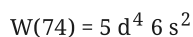
The second transition series also has exceptions.

These are:



There are exceptions in the third transition series as well.

These are:



Because of these exceptions, the electronic configurations of elements in the same group are often dissimilar.

Part (ii)

Conclusion

In each of the three transition series, the number of oxidation states shown by the elements is highest in the middle and lowest at the two ends.

However, +2 and +3 oxidation states are quite stable for all elements in the first transition series — all its metals form stable compounds in the +2 and +3 oxidation states.

The stability of the +2 and +3 oxidation states decreases in the second and third transition series, where higher oxidation states matter more instead.

For example, $[\text{Fe}(\text{Cn})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ are stable complexes, but no such complexes are known for second- and third-series metals such as Mo, W,

Rh, In. They instead form complexes with high oxidation states. For example: WCl_6 , ReF_7 , RuO_4 , etc.

Part (iii)

Conclusion

In each of the three transition series, the first ionisation enthalpy increases from left to right, though there are some exceptions. The first ionisation enthalpies of the third transition series are higher than those of the first and second transition series. This is because of the poor shielding effect of 4 f electrons in the third transition series.

Certain elements in the second transition series have higher first ionisation enthalpies than the elements in the same vertical column of the first transition series. There are also elements in the 2nd transition series whose first ionisation enthalpies are lower than those of the elements in the same vertical column of the 1st transition series.

Part (iv)

Conclusion

Atomic size generally decreases from left to right across a period.

Among the three transition series, the atomic sizes of the elements in the second transition series are greater than those of the elements in the same vertical column of the first transition series.

However, the atomic sizes of the elements in the third transition series are virtually the same as those of the corresponding members in the second transition series.

This is due to lanthanoid contraction.

Q

36

3d electron
count in
hydrated
ions



Write down the number of 3d electrons in each of the following ions: Ti^{2+} , V^{2+} , Cr^{3+} , Mn^{2+} , Fe^{2+} , Fe^{3+} , Co^{2+} , Ni^{2+} and Cu^{2+} . Indicate how would you expect the five 3d orbitals to be occupied for these hydrated ions (octahedral).

Metal ion	Number of d-electrons	Filling of d-orbitals
Ti^{2+}	2	t_{2g}^2
V^{2+}	3	t_{2g}^3
Cr^{3+}	3	t_{2g}^3
Mn^{2+}	5	$t_{2g}^3 e_g^2$
Fe^{2+}	6	$t_{2g}^4 e_g^2$
---	---	---
Fe^{3+}	5	$t_{2g}^3 e_g^2$
Co^{2+}	7	$t_{2g}^5 e_g^2$
Ni^{2+}	8	$t_{2g}^6 e_g^2$
Cu^{2+}	9	$t_{2g}^6 e_g^3$

Q

37

First series' distinct properties



Comment on the statement that elements of the first transition series possess many properties different from those of heavier transition elements.

Conclusion

The properties of the elements of the first transition series differ from those of the heavier transition elements in many ways.

(i) The atomic sizes of the elements of the first transition series are smaller than those of the heavier elements (elements of the 2nd and 3rd transition series). However, the atomic sizes of the elements in the third transition series are virtually the same as those of the corresponding members in the second transition series. This is due to lanthanoid contraction.

(ii) +2 and +3 oxidation states are more common for elements in the first transition series, while higher oxidation states are more common for the heavier elements.

(iii) The enthalpies of atomisation of the elements in the first transition series are lower than those of the corresponding elements in the second and third transition series.

(iv) The melting and boiling points of the first transition series are lower than those of the heavier transition elements.

This is because of stronger metallic bonding (M-M bonding).

(v) The elements of the first transition series form low-spin or high-spin complexes depending upon the strength of the ligand field.

However, the heavier transition elements form only low-spin complexes, irrespective of the strength of the ligand field.

Q

38

Magnetic moment of complex species



What can be inferred from the magnetic moment values of the following complex species ?

Example	Magnetic Moment (BM)
$K_4[Mn(CN)_6]$	2.2
$[Fe(H_2O)_6]^{2+}$	5.3
$K_2[MnCl_4]$	5.9

Key formula

Magnetic

μ) is given as $\mu = \sqrt{n(n+2)}$ moment (.



For value $n = 1$,

$$\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.732$$

For value $n = 2$,

$$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.83$$

For value $n = 3$,

$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87$$

For value $n = 4$,

$$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.899$$

For value $n = 5$,

$$\mu = \sqrt{5(5+2)} = \sqrt{35} = 5.92$$

Substitute

(i) $K_4[Mn(CN)_6]$

In transition metals, the magnetic moment is calculated using the spin-only formula.

So,

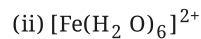
$$\sqrt{n(n+2)} = 2.2$$

This calculated value is closest to $n = 1$. Also, in this complex, Mn is in the +2 oxidation state, so it has 5 electrons in the d orbital.

Conclusion

So we can say that CN^- is a strong field ligand that causes the pairing of electrons.

Substitute



$$\sqrt{n(n+2)} = 5.3$$

This calculated value is closest to $n = 4$. Also, in this complex, Fe is in the +2 oxidation state, so it has 6 electrons in the d orbital.

Conclusion

So we can say that H_2O is a weak field ligand and does not cause the pairing of electrons.

Substitute



$$\sqrt{n(n+2)} = 5.9$$

This calculated value is closest to $n = 5$. Also, in this complex, Mn is in the +2 oxidation state, so it has 5 electrons in the d orbital.

Conclusion

So we can say that Cl^- is a weak field ligand and does not cause the pairing of electrons.